Exact Bayesian Predictive Distributions for the Over-dispersed Poisson Reserving Model with Origin-Period Priors

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Abstract

We derive an exact Bayesian posterior predictive distribution for the over-dispersed Poisson (ODP) reserving model when informative priors are placed on the origin-period parameters. Origin-period parameters are natural for incorporating prior information such as exposures, underwriting changes and external benchmark loss ratios. Bayesian ODP-type models have previously been used to incorporate expert judgement and external information, and study prediction uncertainty, but largely through numerical procedures such as Markov Chain Monte Carlo (MCMC) or approximations. In this paper we show that an exact Bayesian posterior predictive distribution can be written in terms of simple beta and gamma distributions. This yields explicit formulas for reserve means, predictive variances and covariances, together with a simple recursive algorithm for simulating from the full predictive distribution using only standard distributions, which can be used for calculating risk measures such as VaR and TVaR.

We next introduce a one-parameter family of prior distributions which shrink the origin-period parameters toward an exposure-based index. We show that in the strong-prior limit, the model recovers the Cape Cod model, while in the weak-prior limit it recovers the Chain Ladder model. Intermediate prior strength yields a continuum of credibility-weighted estimators between these two endpoints. With a hierarchical prior on the origin period parameters, we show how the blending parameter may itself be estimated from the data. A numerical example illustrates the model naturally weights reserve estimates towards the Cape Cod estimator for more recent origin periods, with increasing weight towards the Chain Ladder estimator for older origin periods.

1 Introduction

The over-dispersed Poisson (ODP) model introduced by Renshaw and Verrall [1] is one of the central stochastic models underlying the chain ladder method. England and Verrall [2] and England and Verrall [3] helped establish bootstrap approaches for assessing reserve uncertainty under the ODP model, which together with Mack's classical reserving model [4] are the workhorses of day-to-day non-life reserve risk modelling which is a crucial input into capital and risk modelling for P&C insurers. Although individual-claim and micro-level reserving models have recently

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attracted substantial academic attention, such models are not yet widely used in practice, with the ODP and Mack's approaches remaining dominant. Bayesian approaches to assessing reserve uncertainty and incorporating expert judgement have been studied in a number of papers over the last two decades. For example, Ntzoufras and Dellaportas [5], Verrall [6], Verrall and England [7] and Verrall [8] are representative of Bayesian reserving work implemented through Markov Chain Monte Carlo (MCMC) or related numerical procedures, including formulations that incorporate expert judgement. Taylor [9] surveys Bayesian chain ladder models more broadly in exponential-dispersion settings.

There is already substantial literature connecting Bayesian ODP-type models to familiar actuarial reserving methods. Wüthrich [10] showed that exponential-dispersion reserving models with conjugate priors lead naturally to credibility-style reserving estimators, and Gisler and Wüthrich [11] showed that credibility estimators for chain ladder are exact Bayesian under exponential-dispersion assumptions with natural conjugate priors. England et al. [12] studied Bayesian ODP models giving rise to classical credibility-type predictors, with uncertainty obtained analytically or numerically.

Recent work has shown that the Bayesian posterior predictive distribution can be analysed exactly when informative priors are placed on the development pattern, yielding a simple recursive representation in terms of beta and gamma distributions and explicit formulas for reserve uncertainty [13].

This paper shows the same is true when priors are placed on the origin-period side. In practice, prior information is often available in origin-period form: exposures, underwriting changes and external benchmark loss ratios all live naturally on the origin side rather than the development side. This is also the natural point of contact between pricing and reserving. Prior and external information connected to origin years has already been studied in a number of forms. Formulas for prediction uncertainty have been obtained for Cape Cod [14] and related credibility-based reserving estimators [15, 16]. Taylor [17] develops a stochastic Cape Cod model within the exponential-dispersion family and also introduces a Bayesian Cape Cod formulation with a MAP estimator. Gigante et al. [18] study hierarchical generalised linear models with origin-year random effects and explicitly allow external information to enter through the random-effect structure. Clark's Bayesian development-pattern work focuses on conjugate priors for the development pattern itself, and explicitly restricts attention to the pattern-selection problem rather than reserve ranges or exposure-based loss-ratio methods [19].

Despite this rich literature, to our knowledge, no previous paper has derived an exact closed-form Bayesian posterior predictive distribution for the ODP model with informative priors placed on the origin-period parameters, together with explicit reserve-uncertainty formulas in that setting. This is the contribution of the present paper.

Aside from academic interest, the practical benefits of an exact closed-form solution are substantial. Exact formulas bring robustness, speed and transparency. They simplify implementation in spreadsheets and production code, make testing much easier, avoid tuning and convergence questions associated with high-dimensional MCMC procedures, and provide a clean analytical benchmark against which simulation-based calculations can be checked. They also give direct structural insight into how prior information and observed data combine.

The present paper fills that gap. We show that, viewing the origin-period parameters as an index of origin-period trends, after integrating out the development parameters the resulting posterior is an inverse-generalised-Dirichlet type family which factorises into independent beta distributions. This produces an exact origin-side conjugate Bayesian ODP model.

The key advantages are as follows. First, the model admits a full posterior predictive distribution in recursive form using only elementary building blocks of independent beta and gamma distributions. Second, reserve means, predictive variances and covariances are available in closed form. Third, under a hierarchical shrinkage prior, the model yields an exact one-parameter credibility bridge between the Chain Ladder and Cape Cod. The strong-prior limit yields Cape Cod, the weak-prior limit yields the data-driven endpoint, and intermediate prior strength yields a continuum of credibility-weighted methods.

An algorithm is presented for generating full simulations of the future claims distribution for internal-model, Solvency II and other capital frameworks. The model projects not only the total reserve but also the full triangle of future payments, allowing discounted cash flow calculations.

The paper is organised as follows. Section 2 introduces the model and derives the origin-side conjugate structure. Section 3 derives the exact posterior predictive distribution, reserve means and reserve-uncertainty formulas. Section 4 introduces the hierarchical shrinkage prior, develops the exact Chain Ladder–Cape Cod bridge. Section 5 records the structure of an empirical illustration. Section 6 concludes.

2 The ODP model under origin-side normalisation

2.1 Model and notation

Consider an incremental triangle $\{X_{ij} : i = 1, \ldots, n,\ j = 1, \ldots, n,\ i + j \leq n + 1\}$ under the ODP model

$$X_{ij} \mid \mu, \beta \sim \phi \operatorname{Poi}\left(\frac{\mu_i \beta_j}{\phi}\right), \qquad i + j \leq n + 1, \tag{1}$$

with conditional independence over observed cells. Here $\phi > 0$ is treated as known, as in England and Verrall [3] and Norman [13]. We keep ϕ constant purely for pedagogical clarity of exposition. As discussed in England and Verrall [3] and Saluz [14], the assumption of a constant scale parameter may not be appropriate in all cases and should be checked. In many cases a scale parameter which varies across development periods is more appropriate. Our results can be extended to that setting, but the constant scale case is sufficient to illustrate the key ideas. A standard limitation of the ODP framework should also be noted: since the cell means must be non-negative, the model is not naturally suited to triangles with material negative incremental entries, for example from salvage and subrogation, or when using incurred/reported claims triangles..

Since the ODP means depend only on the products $\mu_i \beta_j$, any positive rescaling $(\mu, \beta) \mapsto (c\mu, \beta/c)$ leaves the model unchanged. In Norman [13], this invariance was broken by imposing $\sum_j \beta_j = 1$. In this paper we find it more convenient to adopt the constraint

$$\mu_1 = 1, \tag{2}$$

in which case the vector $(\mu_1, \ldots, \mu_n)$ can be interpreted as an index of expected claims by origin period, relative to origin period 1. Fixing $\mu_1 = 1$ selects a convenient representative of an equivalence class rather than changing the substance of the model. Any other positive normalisation would lead to the same fitted cell means and the same predictive distribution after reparameterisation.

Define the scaled incremental and cumulative observations by

$$x_{ij} := \frac{X_{ij}}{\phi}, \qquad d_i := \sum_{j=1}^{n-i+1} x_{ij}, \qquad c_j := \sum_{i=1}^{n-j+1} x_{ij}, \tag{3}$$

and for $m \leq n - i + 1$ define the cumulative scaled claims in origin year i by

$$d_{i,m} := \sum_{j=1}^{m} x_{ij}. \tag{4}$$

The corresponding unscaled observed totals are

$$D_i := \phi d_i = \sum_{j=1}^{n-i+1} X_{ij}, \qquad C_j := \phi c_j = \sum_{i=1}^{n-j+1} X_{ij}. \tag{5}$$

A simple regrouping of the likelihood by development period gives the following factorisation.

Proposition 1. *Under* (1) *and* (2)*, the likelihood kernel is*

$$L(X \mid \mu, \beta) \propto \left(\prod_{i=2}^{n} \mu_i^{d_i} \right) \left(\prod_{j=1}^{n} \beta_j^{c_j} \exp\left[-\frac{\beta_j}{\phi} \sum_{r=1}^{n-j+1} \mu_r \right] \right). \tag{6}$$

Consequently, under the non-informative logarithmic prior

$$\pi(\beta) \propto \prod_{j=1}^{n} \beta_j^{-1}, \tag{7}$$

see Wüthrich [21] for related discussion of non-informative prior choices in Bayesian ODP reserving models. we have, conditional on μ and the observed triangle,

$$\beta_j \mid X, \mu \sim \text{Gamma}\left(c_j, \frac{1}{\phi} \sum_{r=1}^{n-j+1} \mu_r \right), \qquad j = 1, \ldots, n, \tag{8}$$

independently over development periods.

Proof. The likelihood kernel is

$$\prod_{i+j \leq n+1} \left(\frac{\mu_i \beta_j}{\phi} \right)^{x_{ij}} \exp\left(-\frac{\mu_i \beta_j}{\phi} \right), \tag{9}$$

which factorises by collecting the powers of μ_i and β_j. Multiplying by (7) therefore leaves the exponent of β_j equal to $c_j - 1$, giving the stated gamma kernel. □

Integrating out the development parameters first therefore yields the marginal posterior kernel on the origin-period parameters

$$p(\mu \mid X) \propto \pi(\mu) \left(\prod_{i=2}^{n} \mu_i^{d_i} \right) \prod_{j=1}^{n-1} \left(\sum_{r=1}^{n-j+1} \mu_r \right)^{-c_j}, \tag{10}$$

where $\pi(\mu)$ is the prior density on the origin index.

2.2 Generalised inverse-Dirichlet priors on the origin index

A convenient and tractable prior family on the origin index is

$$\pi(\mu) = \prod_{i=2}^{n} \frac{1}{B(a_i, b_i)} \mu_i^{b_i - 1} \left(\sum_{j=1}^{i-1} \mu_j \right)^{a_i} \left(\sum_{j=1}^{i} \mu_j \right)^{-(a_i + b_i)}, \qquad \mu_i > 0, \tag{11}$$

for parameters $a_i > 0$ and $b_i > 0$, $i = 2, \ldots, n$, where $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x + y)$ is the Beta function. We refer to (11) as a generalised inverse-Dirichlet prior [20].

The means, second moments and cross-moments of the μ_i under (11) are for $i \geq 2$,

$$\mathbb{E}[\mu_i] = \frac{b_i}{a_i - 1} \prod_{k=2}^{i-1} \frac{a_k + b_k - 1}{a_k - 1}, \qquad a_k > 1 \text{ for } k = 2, \ldots, i, \tag{12}$$

$$\mathbb{E}[\mu_i^2] = \frac{b_i(a_i + b_i - 1)}{(a_i - 1)(a_i - 2)} \left(\prod_{k=2}^{i-1} \frac{(a_k + b_k - 1)(a_k + b_k - 2)}{(a_k - 1)(a_k - 2)} \right) \left(\prod_{k=i+1}^{i-1} \frac{a_k + b_k - 1}{a_k - 1} \right) \frac{b_i}{a_i - 1}, \tag{13}$$

and for $2 \leq i < j \leq n$,

$$\mathbb{E}[\mu_i \mu_j] = \frac{b_i(a_i + b_i - 1)}{(a_i - 1)(a_i - 2)} \left(\prod_{k=2}^{i-1} \frac{(a_k + b_k - 1)(a_k + b_k - 2)}{(a_k - 1)(a_k - 2)} \right) \left(\prod_{k=i+1}^{j-1} \frac{a_k + b_k - 1}{a_k - 1} \right) \frac{b_j}{a_j - 1}, \tag{14}$$

provided $a_k > 2$ for $k = 2, \ldots, i$ and $a_k > 1$ for $k = i + 1, \ldots, j$. The covariance between μ_i and μ_j is then easily obtained:

$$\mathrm{Cov}(\mu_i, \mu_j) = \mathbb{E}[\mu_i \mu_j] - \mathbb{E}[\mu_i]\mathbb{E}[\mu_j]. \tag{15}$$

Using equations (12) and (15), one can calibrate the parameters a_i and b_i to match desired prior means and variances for the origin index μ.

A property of the generalised inverse-Dirichlet family is that the prior density factorises into a product of independent beta densities after a convenient change of coordinates. This is the key to the exact conjugacy and tractability of the model. Setting

$$\xi_i := \frac{\sum_{j=1}^{i-1} \mu_j}{\sum_{j=1}^{i} \mu_j} \in (0, 1), \qquad i = 2, \ldots, n. \tag{16}$$

Then

$$\sum_{j=1}^{i} \mu_j = \prod_{k=2}^{i} \xi_k^{-1}, \qquad \mu_i = \frac{1-\xi_i}{\xi_i} \prod_{k=2}^{i-1} \xi_k^{-1}, \qquad i = 2, \ldots, n. \tag{17}$$

The role of the transformation (16) is that it converts the generalised inverse-Dirichlet prior into a product of independent beta laws.

Proposition 2. *Under the transformation* (16)*, the prior* (11) *is equivalent to*

$$\xi_i \overset{\text{ind}}{\sim} \text{Beta}(a_i, b_i), \qquad i = 2, \ldots, n. \tag{18}$$

Conversely, any independent beta prior of the form (18) *induces the density* (11) *on* μ*.*

Proof. Since $\xi_i = \sum_{j=1}^{i-1} \mu_j / \sum_{j=1}^{i} \mu_j$, we have $1 - \xi_i = \mu_i / \sum_{j=1}^{i} \mu_j$. The map

$$(\mu_2, \ldots, \mu_n) \mapsto (\xi_2, \ldots, \xi_n)$$

is triangular, with Jacobian

$$\left| \frac{\partial(\xi_2, \ldots, \xi_n)}{\partial(\mu_2, \ldots, \mu_n)} \right| = \prod_{i=2}^{n} \frac{\sum_{j=1}^{i-1} \mu_j}{\sum_{j=1}^{i} \mu_j}^2.$$

Therefore

$$\prod_{i=2}^{n} \frac{1}{B(a_i, b_i)} \xi_i^{a_i - 1} (1 - \xi_i)^{b_i - 1} \left| \frac{\partial \xi}{\partial \mu} \right| = \prod_{i=2}^{n} \frac{1}{B(a_i, b_i)} \mu_i^{b_i - 1} \left(\sum_{j=1}^{i-1} \mu_j \right)^{a_i} \left(\sum_{j=1}^{i} \mu_j \right)^{-(a_i + b_i)},$$

which is exactly (11). □

For later use, define for $i = 2, \ldots, n$

$$s_i := \sum_{k=1}^{i-1} d_{k,n-i+1}. \tag{19}$$

Under the logarithmic development prior (7), integrating out the development parameters yields

$$p(\xi \mid X) \propto \pi_\xi(\xi) \prod_{i=2}^{n} \xi_i^{s_i} (1 - \xi_i)^{d_i}, \tag{20}$$

where π_ξ denotes the prior density induced on the transformed coordinates. In particular, the family (11) is precisely the class of origin-side priors for which (20) factorises into independent beta terms.

3 Exact predictive distribution and reserve uncertainty

This section derives the exact posterior predictive distribution and reserve-uncertainty formulas under the logarithmic development prior (7) and the generalised inverse-Dirichlet origin prior (11).

3.1 General posterior under independent beta priors on the transformed origin coordinates

By Proposition 2, the prior (11) is equivalent to the independent beta specification

$$\xi_i \overset{\text{ind}}{\sim} \text{Beta}(a_i, b_i), \qquad i = 2, \ldots, n.$$

Combining this with the kernel (20) gives an exact conjugate posterior.

Theorem 3. *Under* (7) *and* (11),

$$\xi_i \mid X \overset{\text{ind}}{\sim} \text{Beta}\big(a_i + s_i,\, b_i + d_i\big), \qquad i = 2, \ldots, n. \tag{21}$$

Proof. Multiply the beta prior density in the ξ coordinates by the kernel (20) and collect powers of ξ_i and $(1 - \xi_i)$. □

The posterior inverse moments needed for reserving are therefore

$$\mathbb{E}\left[\frac{1-\xi_i}{\xi_i}\,\middle|\, X\right] = \frac{b_i + d_i}{a_i + s_i - 1}, \qquad \mathbb{E}\big[\xi_i^{-1} \,\big|\, X\big] = \frac{a_i + b_i + s_i + d_i - 1}{a_i + s_i - 1}, \tag{22}$$

and, when the moments exist,

$$\mathbb{E}\left[\left(\frac{1-\xi_i}{\xi_i}\right)^2\,\middle|\, X\right] = \frac{(b_i + d_i)(b_i + d_i + 1)}{(a_i + s_i - 1)(a_i + s_i - 2)}, \tag{23}$$

$$\mathbb{E}\big[\xi_i^{-2} \,\big|\, X\big] = \frac{(a_i + b_i + s_i + d_i - 1)(a_i + b_i + s_i + d_i - 2)}{(a_i + s_i - 1)(a_i + s_i - 2)}. \tag{24}$$

3.2 Posterior predictive simulation under the general prior

For development period j, the future cells are those with $i = n - j + 2, \ldots, n$. Conditional on X and ξ, the development parameters remain independent with

$$\beta_j \mid X, \xi \sim \text{Gamma}\left(c_j, \frac{1}{\phi}\sum_{r=1}^{n-j+1} \mu_r\right), \qquad j = 1, \ldots, n, \tag{25}$$

by (8). Hence future payments may be simulated exactly as follows:

1. For $i = 2, \ldots, n$, sample ξ_i independently from (21).
2. Recover μ_i from ξ using (17).
3. For $j = 2, \ldots, n$, sample β_j independently from (25).
4. For each future cell (i, j) with $i + j > n + 1$, sample

$$X_{ij}^{\text{fut}} \mid X, \mu, \beta \sim \phi\,\text{Poi}\left(\frac{\mu_i \beta_j}{\phi}\right).$$

This representation uses only independent beta draws, independent gamma draws, and conditionally independent scaled Poisson draws. For the final forecasting step, one may prefer to replace the scaled Poisson draw with $X_{ij}^{\text{fut}} \mid X, \mu, \beta \sim \phi \,\text{Gamma}\Big(\frac{\mu_i \beta_j}{\phi}, 1\Big)$. which has the same mean and variance but avoids the discreteness inherent in the Poisson distribution.

3.3 Reserve mean under the general prior

For a future cell (i, j) with $i + j > n + 1$, define

$$W_{ij} := \frac{\mu_i}{\sum_{r=1}^{n-j+1} \mu_r} = \frac{1 - \xi_i}{\xi_i} \prod_{k=n-j+2}^{i-1} \xi_k^{-1}. \tag{26}$$

Then $\mathbb{E}[X_{ij} \mid X, \xi] = C_j W_{ij}$, since $\mathbb{E}[\beta_j \mid X, \xi] = C_j / \sum_{r=1}^{n-j+1} \mu_r$.

Proposition 4. *For a future cell (i, j), define*

$$m_{ij} := \mathbb{E}[W_{ij} \mid X]. \tag{27}$$

Then

$$m_{ij} = \frac{b_i + d_i}{a_i + s_i - 1} \prod_{k=n-j+2}^{i-1} \frac{a_k + b_k + s_k + d_k - 1}{a_k + s_k - 1}, \tag{28}$$

provided the displayed expectations exist.

Proof. Use (22) together with independence of the ξ_k. □

Hence the posterior mean reserve for origin period i is

$$\widehat{R}_i := \mathbb{E}\left[\sum_{j=n-i+2}^{n} X_{ij} \,\middle|\, X \right] = \sum_{j=n-i+2}^{n} C_j m_{ij}, \tag{29}$$

and the total reserve mean is

$$\widehat{R} := \mathbb{E}[T \mid X] = \sum_{i+j>n+1} C_j m_{ij}, \qquad T := \sum_{i+j>n+1} X_{ij}. \tag{30}$$

3.4 Reserve variances and covariances under the general prior

For a future cell, define

$$m_{ij}^{(2)} := \mathbb{E}[W_{ij}^2 \mid X]. \tag{31}$$

Then

$$m_{ij}^{(2)} = \frac{(b_i + d_i)(b_i + d_i + 1)}{(a_i + s_i - 1)(a_i + s_i - 2)} \prod_{k=n-j+2}^{i-1} \frac{(a_k + b_k + s_k + d_k - 1)(a_k + b_k + s_k + d_k - 2)}{(a_k + s_k - 1)(a_k + s_k - 2)}. \tag{32}$$

For mixed moments, define for any finite set $\mathcal{S}$ of future cells the exponents $a_k(\mathcal{S})$ and $b_k(\mathcal{S})$

by

$$\prod_{(i,j)\in\mathcal{S}} W_{ij} = \prod_{k=2}^{n} \xi_k^{-a_k(\mathcal{S})}(1-\xi_k)^{b_k(\mathcal{S})}. \tag{33}$$

Then

$$\mathbb{E}\left[\prod_{(i,j)\in\mathcal{S}} W_{ij} \,\middle|\, X\right] = \prod_{k=2}^{n} \frac{B\big(a_k+s_k-a_k(\mathcal{S}),\, b_k+d_k+b_k(\mathcal{S})\big)}{B\big(a_k+s_k,\, b_k+d_k\big)}, \tag{34}$$

provided the moments exist. In particular, for two future cells (i,j) and (ℓ,h), define

$$q_{ij,\ell h} := \mathbb{E}[W_{ij}W_{\ell h} \mid X]. \tag{35}$$

Proposition 5. *For each future cell (i,j),*

$$\mathrm{Var}(X_{ij} \mid X) = \phi C_j m_{ij} + \phi C_j m_{ij}^{(2)} + C_j^2\big(m_{ij}^{(2)} - m_{ij}^2\big). \tag{36}$$

For two distinct future cells,

$$\mathrm{Cov}(X_{ij}, X_{\ell j} \mid X) = \phi C_j q_{ij,\ell j} + C_j^2\big(q_{ij,\ell j} - m_{ij}m_{\ell j}\big), \qquad i \neq \ell, \tag{37}$$

$$\mathrm{Cov}(X_{ij}, X_{\ell h} \mid X) = C_j C_h\big(q_{ij,\ell h} - m_{ij}m_{\ell h}\big), \qquad j \neq h. \tag{38}$$

Proof. Conditional on ξ, the development periods are independent, $\mathbb{E}[X_{ij} \mid X,\xi] = C_j W_{ij}$, and

$$\mathrm{Var}(X_{ij} \mid X,\xi) = \phi C_j W_{ij} + \phi C_j W_{ij}^2.$$

The same-column covariance given ξ is

$$\mathrm{Cov}(X_{ij}, X_{\ell j} \mid X,\xi) = \phi C_j W_{ij}W_{\ell j}, \qquad i \neq \ell,$$

while different development periods are conditionally independent. Applying the law of total variance and the law of total covariance then gives the result. □

To simplify the calculation of reserve variances efficiently without relying on individual pairwise cell covariances, it is helpful to exploit the independence of the posterior beta coordinates and the telescoping structure of (26). For $k = 2,\dots,n$, define the posterior inverse moments

$$v_k := \mathbb{E}[\xi_k^{-1} \mid X] = \frac{a_k+b_k+s_k+d_k-1}{a_k+s_k-1} \tag{39}$$

and

$$v_k^{(2)} := \mathbb{E}[\xi_k^{-2} \mid X] = \frac{(a_k+b_k+s_k+d_k-1)(a_k+b_k+s_k+d_k-2)}{(a_k+s_k-1)(a_k+s_k-2)}. \tag{40}$$

Then the first and second moments of the future-cell weights can be written as

$$m_{ij} = \left(\frac{b_i+d_i}{a_i+s_i-1}\right) \prod_{k=n-j+2}^{i-1} v_k \tag{41}$$

and

$$m_{ij}^{(2)} = \left(\frac{(b_i + d_i)(b_i + d_i + 1)}{(a_i + s_i - 1)(a_i + s_i - 2)} \right) \prod_{k=n-j+2}^{i-1} v_k^{(2)}. \tag{42}$$

By applying the law of total variance conditional on ξ, the variances of the origin-period reserves and the total reserve collapse into sums over future development periods.

Proposition 6 (Origin-period reserve variance)**.** *For a given origin period i, let*

$$T_i^{\text{fut}} := \sum_{j=n-i+2}^{n} X_{ij}. \tag{43}$$

Then

$$\operatorname{Var}(T_i^{\text{fut}} \mid X) = \sum_{j=n-i+2}^{n} \phi C_j \big(m_{ij} + m_{ij}^{(2)}\big) + \sum_{j=n-i+2}^{n} \sum_{h=n-i+2}^{n} C_j C_h \Sigma_{jh}^{(i)}, \tag{44}$$

where, for $j \le h$,

$$\Sigma_{jh}^{(i)} := m_{ih} \left(\frac{m_{ij}^{(2)}}{m_{ij}} - m_{ij} \right). \tag{45}$$

By symmetry, $\Sigma_{jh}^{(i)} = \Sigma_{hj}^{(i)}$ for $j > h$.

Proof. Conditional on ξ, the cells X_{ij} in row i are independent because they depend on independent development parameters β_j. Hence

$$\operatorname{Var}(T_i^{\text{fut}} \mid X) = \mathbb{E}\big[\operatorname{Var}(T_i^{\text{fut}} \mid X, \xi) \mid X\big] + \operatorname{Var}\big(\mathbb{E}[T_i^{\text{fut}} \mid X, \xi] \mid X\big). \tag{46}$$

The conditional process-variance term gives the first sum in (44). For the second term, if $j \le h$ then

$$W_{ih} = W_{ij} \prod_{k=n-h+2}^{n-j+1} \xi_k^{-1}, \tag{47}$$

and the extra product is independent of W_{ij}. Therefore

$$\mathbb{E}[W_{ij} W_{ih} \mid X] = m_{ij}^{(2)} \frac{m_{ih}}{m_{ij}}, \tag{48}$$

which yields (45) after subtracting $m_{ij} m_{ih}$. □

For the total reserve variance, the same telescoping structure leads to an even simpler column-based representation.

Proposition 7 (Total reserve mean and variance)**.** *The total reserve*

$$T := \sum_{i+j>n+1} X_{ij} \tag{49}$$

has posterior predictive mean

$$\mathbb{E}[T \mid X] = \sum_{j=2}^{n} C_j \Big(\prod_{k=n-j+2}^{n} v_k - 1 \Big) \tag{50}$$

and posterior predictive variance

$$\mathrm{Var}(T \mid X) = \sum_{j=2}^{n} \phi C_j \Big(\prod_{k=n-j+2}^{n} v_k^{(2)} - \prod_{k=n-j+2}^{n} v_k \Big) + \sum_{j=2}^{n} \sum_{h=2}^{n} C_j C_h \Omega_{jh}, \tag{51}$$

where, for $j \le h$,

$$\Omega_{jh} := \prod_{k=n-h+2}^{n} v_k \left(\frac{\prod_{k=n-j+2}^{n} v_k^{(2)}}{\prod_{k=n-j+2}^{n} v_k} - \prod_{k=n-j+2}^{n} v_k \right). \tag{52}$$

By symmetry, $\Omega_{jh} = \Omega_{hj}$ *for* $j > h$.

Proof. Let

$$S_j := \sum_{i=n-j+2}^{n} W_{ij} \tag{53}$$

denote the future proportion of development period j. Substituting (17) into (26), the sum telescopes to

$$S_j = \left(\prod_{k=n-j+2}^{n} \xi_k^{-1} \right) - 1. \tag{54}$$

Hence, with

$$P_j := S_j + 1 = \prod_{k=n-j+2}^{n} \xi_k^{-1}, \tag{55}$$

we have $\mathbb{E}[P_j \mid X] = \Pi_j$ and $\mathbb{E}[P_j^2 \mid X] = \Pi_j^{(2)}$. Since

$$\mathbb{E}[T \mid X, \xi] = \sum_{j=2}^{n} C_j S_j, \tag{56}$$

(50) follows immediately. Also, conditional on ξ, the future total in development period j has variance

$$\phi C_j \big(S_j + S_j^2 \big), \tag{57}$$

so taking expectations gives the first sum in (51). For the covariance term, if $j \le h$ then

$$P_h = P_j \prod_{k=n-h+2}^{n-j+1} \xi_k^{-1}, \tag{58}$$

with the extra product independent of P_j, so

$$\mathbb{E}[P_j P_h \mid X] = \Pi_h \frac{\Pi_j^{(2)}}{\Pi_j}. \tag{59}$$

Subtracting $\mathbb{E}[P_j \mid X]\mathbb{E}[P_h \mid X] = \Pi_j \Pi_h$ gives (52). Applying the law of total variance yields (51). □

4 An exposure-centred one-parameter family and its classical limits

The previous sections derived the exact predictive distribution under a generalised inverse Dirichlet origin-side prior. We now consider the special case of a prior where the expected claims by origin period are proportional to the exposure. A single concentration parameter then controls how tightly the prior is concentrated around the exposure.

We will show that in the weak-prior limit - that is, when the prior variance goes to infinity, the model recovers the Chain Ladder mean. The strong-prior limit in which the prior expected claims are exactly proportional to the exposure recovers the Cape Cod mean. Thus the family provides a continuous interpolation between these two classical methods, with the concentration parameter controlling the strength of the credibility effect. The family is also convenient for practical use, since it has only one additional prior parameter to calibrate. The concentration parameter can be estimated from the data through maximum marginal likelihood. Introducing a hyperprior on the concentration parameter then gives a fully Bayesian model for the concentration parameter.

4.1 The exposure-centred one-parameter prior family

Define the relative exposure index by

$$e_i := \frac{E_i}{E_1}, \qquad i = 1, \dots, n, \qquad e_1 = 1. \tag{60}$$

For $\tau \geq 0$, define the prior on the origin-period parameters by the generalised inverse-Dirichlet family (11) with parameters given by

$$a_i = 1 + \tau \sum_{r=1}^{i-1} e_r, \qquad b_i = \tau e_i. \tag{61}$$

Equivalently, in the transformed coordinates,

$$\xi_i \mid \tau \overset{\text{ind}}{\sim} \text{Beta}\left(1 + \tau \sum_{r=1}^{i-1} e_r,\, \tau e_i\right), \qquad i = 2, \dots, n. \tag{62}$$

This family has the appealing property that the prior mean of μ_i is proportional to the exposure E_i. From (12),

$$\mathbb{E}[\mu_i \mid \tau] = e_i = \frac{E_i}{E_1}, \qquad i = 2, \dots, n. \tag{63}$$

The parameter τ controls how strongly the prior is concentrated around the exposure. Smaller τ corresponds to a weaker prior, while larger τ corresponds to a stronger prior. We will see that the weak prior limit $\tau \to 0$ leads to the Chain Ladder endpoint; the strong prior limit $\tau \to \infty$ is the Cape Cod endpoint. A construction that gives rise to this prior using independent gamma prior loss ratios is given in Appendix A.

Under (21) and (61),

$$\xi_i \mid X, \tau \overset{\text{ind}}{\sim} \text{Beta}\big(u_i(\tau), v_i(\tau)\big), \qquad i = 2, \dots, n, \tag{64}$$

with

$$u_i(\tau) := 1 + s_i + \tau \sum_{r=1}^{i-1} e_r, \qquad v_i(\tau) := d_i + \tau e_i. \tag{65}$$

The credibility effect is clearly visible in the posterior mean. Defining

$$\xi_i^0 := \frac{\sum_{r=1}^{i-1} e_r}{\sum_{r=1}^{i} e_r}, \qquad i = 2, \dots, n. \tag{66}$$

the posterior mean of each transformed coordinate is

$$\mathbb{E}[\xi_i \mid X, \tau] = \frac{1 + s_i + \tau \sum_{r=1}^{i-1} e_r}{1 + s_i + d_i + \tau \sum_{r=1}^{i} e_r} = z_i(\tau)\, \xi_i^0 + \big(1 - z_i(\tau)\big) \widehat{\xi}_i^{\mathrm{CL}}, \tag{67}$$

where

$$\widehat{\xi}_i^{\mathrm{CL}} := \frac{1 + s_i}{1 + s_i + d_i}, \qquad z_i(\tau) := \frac{\tau \sum_{r=1}^{i} e_r}{1 + s_i + d_i + \tau \sum_{r=1}^{i} e_r}. \tag{68}$$

Likewise,

$$\mathbb{E}\big[\xi_i^{-1} \,\big|\, X, \tau\big] = \frac{s_i + d_i + \tau \sum_{r=1}^{i} e_r}{s_i + \tau \sum_{r=1}^{i-1} e_r} = w_i(\tau)\, \frac{s_i + d_i}{s_i} + \big(1 - w_i(\tau)\big) \frac{1}{\xi_i^0}, \tag{69}$$

with

$$w_i(\tau) := \frac{s_i}{s_i + \tau \sum_{r=1}^{i-1} e_r}. \tag{70}$$

For fixed τ, reserve means and predictive variances are obtained by substituting (65) into the general formulas of Section 3. The next two subsections verify the two classical endpoints, starting with the fully data-driven Chain Ladder limit and then moving to the exposure-driven Cape Cod limit.

4.2 The Chain Ladder weak-prior limit

As $\tau \to 0$ in (61), the exposure-centred family tends to an improper prior. In ξ coordinates, the limit is

$$\xi_i \sim \mathrm{Beta}(1, 0), \qquad i = 2, \dots, n, \tag{71}$$

Then Theorem 3 gives

$$\xi_i \mid X \sim \mathrm{Beta}(1 + s_i, d_i), \qquad i = 2, \dots, n. \tag{72}$$

Consequently,

$$\mathbb{E}\left[\frac{1 - \xi_i}{\xi_i} \,\middle|\, X\right] = \frac{d_i}{s_i}, \qquad \mathbb{E}\big[\xi_i^{-1} \,\big|\, X\big] = \frac{s_i + d_i}{s_i}. \tag{73}$$

Substituting into (28) gives

$$\mathbb{E}[X_{ij} \mid X] = C_j \frac{d_i}{s_i} \prod_{k=n-j+2}^{i-1} \frac{s_k + d_k}{s_k}. \tag{74}$$

The next proposition shows that it coincides exactly with the usual development-side Chain Ladder completion.

Proposition 8 (Transpose invariance of Chain Ladder)**.** *Let*

$$D_{ij} := \sum_{k=1}^{j} X_{ik}, \qquad i + j \le n + 1, \tag{75}$$

be the development-cumulative triangle, with Chain Ladder factors

$$f_j := \frac{\sum_{i=1}^{n-j} D_{i,j+1}}{\sum_{i=1}^{n-j} D_{i,j}}, \qquad j = 1, \dots, n-1. \tag{76}$$

Also let

$$O_{ij} := \sum_{k=1}^{i} X_{kj}, \qquad i + j \le n + 1, \tag{77}$$

be the origin-cumulative triangle, with factors obtained by applying Chain Ladder to the transposed triangle,

$$g_i := \frac{\sum_{j=1}^{n-i} O_{i+1,j}}{\sum_{j=1}^{n-i} O_{i,j}}, \qquad i = 1, \dots, n-1. \tag{78}$$

Then both procedures produce the same completed incremental triangle.

Proof. Define

$$q_j := \prod_{k=j}^{n-1} f_k^{-1}, \qquad q_n = 1, \qquad q_0 := 0, \tag{79}$$

and set

$$p_j := q_j - q_{j-1}, \qquad j = 1, \dots, n. \tag{80}$$

Ordinary Chain Ladder then gives the completed cumulative triangle

$$\widehat{D}_{ij} = \widehat{U}_i q_j, \qquad \widehat{U}_i = \frac{D_{i,n-i+1}}{q_{n-i+1}}, \tag{81}$$

so the completed incremental triangle is

$$\widehat{X}_{ij} = \widehat{D}_{ij} - \widehat{D}_{i,j-1} = \widehat{U}_i p_j. \tag{82}$$

Now define the fitted origin-cumulative triangle

$$\widehat{O}_{ij} := \sum_{k=1}^{i} \widehat{X}_{kj} = p_j \sum_{k=1}^{i} \widehat{U}_k =: p_j V_i. \tag{83}$$

For observed cells with $i + j \le n + 1$, every summand in $O_{ij} = \sum_{k=1}^{i} X_{kj}$ is observed, so $\widehat{O}_{ij} = O_{ij}$ on the observed region. Therefore

$$g_i = \frac{\sum_{j=1}^{n-i} O_{i+1,j}}{\sum_{j=1}^{n-i} O_{i,j}} = \frac{\sum_{j=1}^{n-i} \widehat{O}_{i+1,j}}{\sum_{j=1}^{n-i} \widehat{O}_{i,j}} = \frac{V_{i+1}}{V_i}. \tag{84}$$

For fixed development period j, transposed Chain Ladder yields

$$\widetilde{O}_{ij} = O_{n-j+1,j} \prod_{k=n-j+1}^{i-1} g_k, \qquad i > n-j+1. \tag{85}$$

Since $O_{n-j+1,j} = \widehat{O}_{n-j+1,j} = p_j V_{n-j+1}$ and $g_k = V_{k+1}/V_k$,

$$\widetilde{O}_{ij} = p_j V_{n-j+1} \prod_{k=n-j+1}^{i-1} \frac{V_{k+1}}{V_k} = p_j V_i = \widehat{O}_{ij}. \tag{86}$$

Taking first differences in the origin direction gives

$$\widetilde{X}_{ij} = \widetilde{O}_{ij} - \widetilde{O}_{i-1,j} = p_j(V_i - V_{i-1}) = \widehat{U}_i p_j = \widehat{X}_{ij}. \tag{87}$$

Thus the completed incremental triangles coincide. □

Corollary 9. *The weak-prior limit* (71) *recovers the classical Chain Ladder mean exactly.*

4.3 Cape Cod as the strong-prior exposure-centred limit

At the other endpoint, as $\tau \to \infty$ in (61), the origin-period prior concentrates on the exposure index, so that

$$\mu_i \to e_i = \frac{E_i}{E_1},$$

In this limit, (8) gives

$$\mathbb{E}[\beta_j \mid X] = \phi \frac{c_j}{\sum_{r=1}^{n-j+1} \mu_r^0} = E_1 \frac{C_j}{\sum_{r=1}^{n-j+1} E_r}. \tag{88}$$

Define the additive loss-ratio components

$$\ell_j := \frac{\beta_j}{E_1}, \qquad j = 1, \ldots, n. \tag{89}$$

Then

$$\mathbb{E}[\ell_j \mid X] = \frac{C_j}{\sum_{r=1}^{n-j+1} E_r}. \tag{90}$$

Summing over development periods gives the common loss ratio

$$\widehat{\kappa}_{\mathrm{CC}} := \sum_{j=1}^{n} \frac{C_j}{\sum_{r=1}^{n-j+1} E_r}. \tag{91}$$

Hence the posterior expected ultimate in origin year i is

$$\widehat{U}_i^{\mathrm{CC}} = \mathbb{E}\left[\sum_{j=1}^{n} \mu_i \beta_j \,\middle|\, X \right] = E_i \widehat{\kappa}_{\mathrm{CC}}. \tag{92}$$

Define the cumulative paid proportion by

$$\widehat{q}_m^{\mathrm{CC}} = \frac{\sum_{j=1}^{m} C_j / \sum_{r=1}^{n-j+1} E_r}{\sum_{j=1}^{n} C_j / \sum_{r=1}^{n-j+1} E_r}, \qquad m = 1, \ldots, n. \tag{93}$$

Then

$$\sum_{i=1}^{n} E_i \widehat{q}_{n-i+1}^{\mathrm{CC}} = \frac{\sum_{j=1}^{n} C_j}{\widehat{\kappa}_{\mathrm{CC}}}, \tag{94}$$

so that

$$\widehat{\kappa}_{\mathrm{CC}} = \frac{\sum_{j=1}^{n} C_j}{\sum_{i=1}^{n} E_i \widehat{q}_{n-i+1}^{\mathrm{CC}}}. \tag{95}$$

Therefore

$$\widehat{R}_i^{\mathrm{CC}} = E_i \widehat{\kappa}_{\mathrm{CC}} \big(1 - \widehat{q}_{n-i+1}^{\mathrm{CC}}\big). \tag{96}$$

Thus the strong-prior exposure-centred limit of the exact Bayesian model recovers the classical Cape Cod common loss ratio, additive loss-ratio profile and reserve formula exactly. The word 'strong' here refers to the origin-period index: the relative ultimate claims by origin period are fixed in proportion to exposure, while the common loss-ratio level is still driven from the triangle.

4.4 Adding a hyperprior on the shrinkage strength

The hierarchical model is introduced to express a familiar actuarial idea: origin-year expected loss ratios are usually similar, but not identical. Pure Chain Ladder estimates each origin period from its own claims experience with essentially no pooling, while pure Cape Cod imposes a common exposure-scaled loss ratio profile.

The fixed-τ family already gives an exact credibility bridge between Chain Ladder and Cape Cod. The marginal likelihood is

$$L_m(\tau; X) \propto \prod_{i=2}^{n} \frac{B\big(u_i(\tau), v_i(\tau)\big)}{B\Big(1 + \tau \sum_{r=1}^{i-1} e_r, \tau e_i\Big)}, \tag{97}$$

One may also place a hyperprior $\pi(\tau)$ on τ. The resulting posterior is:

$$p(\tau \mid X) \propto \pi(\tau) L_m(\tau; X). \tag{98}$$

This gives a simple one-dimensional route to several implementations. In a fully Bayesian treatment one may integrate over $p(\tau \mid X)$ by quadrature or simulation. In this case it is important to ensure that the prior $\pi(\tau)$ is proper, since the marginal likelihood is finite as $\tau \to \infty$. In an empirical Bayes treatment one may maximise the marginal likelihood to obtain a maximum marginal likelihood estimate τ^{MML}. Alternatively, with a proper hyperprior $\pi(\tau)$, one may maximise (98) to obtain the MAP estimate τ^{MAP}.

Once a value of τ has been selected, either by maximisation or by simulation from the hyperposterior, all reserve moments and simulation formulas below apply without further approximation.

The fully Bayesian prediction error is then

$$\mathrm{Var}(T \mid X) = \int_0^\infty \big(V(\tau) + \widehat{R}(\tau)^2\big) p(\tau \mid X)\, \mathrm{d}\tau - \widehat{R}^2, \tag{99}$$

where $V(\tau)$ denotes the fixed-τ posterior predictive variance of T and

$$\widehat{R} := \int_0^\infty \widehat{R}(\tau) p(\tau \mid X)\, \mathrm{d}\tau. \tag{100}$$

5 Illustrative numerical example

To illustrate the method we use the numerical example from Saluz [14]. This is a particularly useful benchmark because Saluz [14] report the classical Cape Cod results, together with approximate reserve standard deviations based on maximum likelihood estimates on the same data. The example therefore provides a direct numerical comparison between the classical Cape Cod estimator, the exact data-driven ODP or Chain Ladder endpoint, and the new hierarchical model derived in this paper. It also makes the practical interpretation of the model transparent: the exposure benchmark does not replace the data, but acts as a structured prior that stabilises the more immature origin-year estimates while leaving the mature part of the triangle essentially unchanged. Table 1 gives the incremental payments and Table 2 gives the premiums and implied exposure-based index for the exposure-centred prior. Note that here we have not adjusted the historical premiums for rate changes and other factors, which would be necessary in practice to obtain a more realistic exposure benchmark.

Origin i	1	2	3	4	5	6	7	8	9	10
1	5,946,975	3,721,237	895,717	207,761	206,704	62,124	65,813	14,850	11,129	15,814
2	6,346,756	3,246,406	723,221	151,797	67,824	36,604	52,752	11,186	11,646	
3	6,269,090	2,976,223	847,053	262,768	152,703	65,445	53,545	8,924		
4	5,863,015	2,683,224	722,532	190,653	132,975	88,341	43,328			
5	5,778,885	2,745,229	653,895	273,395	230,288	105,224				
6	6,184,793	2,828,339	572,765	244,899	104,957					
7	5,600,184	2,893,207	563,114	225,517						
8	5,288,066	2,440,103	528,042							
9	5,290,793	2,357,936								
10	5,675,568									

Table 1: Incremental payments and premiums from the Saluz Cape Cod example.

For the numerical illustration of the hierarchical model we use the plug-in estimate of τ obtained by maximising the marginal likelihood in (98), rather than averaging over a hyperposterior for τ. This gives

$$\widehat{\tau}_{\mathrm{ML}} \approx 346. \tag{101}$$

Figure 1 shows the normalised marginal likelihood as a function of τ. The maximum is quite well-defined. It is also interesting to note that the marginal likelihood is quite flat for large τ. In fact the marginal likelihood is finite as $\tau \to \infty$, so the posterior is improper unless a proper prior is placed on τ.

Using the selected ODP scale parameter $\phi = 21,610.84$, estimated from the Cape Cod fitted triangle as in Saluz [14], the resulting posterior mean reserves and reserve standard deviations

Origin i	Premium E_i	Relative premium E_i/E_1
1	15,473,558	1.000000
2	14,882,436	0.961798
3	14,456,039	0.934241
4	14,054,917	0.908318
5	14,525,373	0.938722
6	15,025,923	0.971071
7	14,832,965	0.958601
8	14,550,359	0.940337
9	14,461,781	0.934613
10	15,210,363	0.982991

Table 2: Premiums and exposure benchmark index implied by the premiums.

	Chain Ladder		Hierarchical		Cape Cod	
Origin	Reserve	LR	Reserve	LR	Reserve	LR
Origin 1	0	0.7205	0	0.7205	0	0.7205
Origin 2	15,126	0.7165	15,159	0.7165	15,209	0.7165
Origin 3	26,257	0.7375	26,000	0.7375	25,619	0.7375
Origin 4	34,538	0.6943	35,075	0.6944	35,874	0.6944
Origin 5	85,302	0.6797	87,294	0.6798	90,234	0.6800
Origin 6	156,494	0.6717	160,597	0.6719	166,584	0.6723
Origin 7	286,121	0.6451	297,878	0.6459	314,665	0.6470
Origin 8	449,167	0.5983	482,247	0.6006	528,056	0.6037
Origin 9	1,043,242	0.6010	1,112,597	0.6058	1,200,821	0.6119
Origin 10	3,950,815	0.6329	4,105,566	0.6431	4,240,563	0.6519
Total	6,047,061	0.6699	6,322,415	0.6717	6,617,625	0.6737

Table 3: Posterior mean reserves and ultimate loss ratios by origin year, using the exact analytic approach.

	Chain Ladder		Hierarchical		Cape Cod	
Origin	SD	CoV (%)	SD	CoV (%)	SD	CoV (%)
Origin 1	0		0		0	
Origin 2	25,332	167.47	25,356	167.27	25,393	166.96
Origin 3	31,647	120.53	31,403	120.78	31,041	121.16
Origin 4	34,357	99.48	34,686	98.89	35,172	98.04
Origin 5	50,589	59.31	51,251	58.71	52,218	57.87
Origin 6	66,816	42.70	67,696	42.15	68,958	41.40
Origin 7	88,204	30.83	89,814	30.15	92,035	29.25
Origin 8	109,265	24.33	112,633	23.36	116,992	22.16
Origin 9	170,259	16.32	171,871	15.45	173,182	14.42
Origin 10	401,963	10.17	361,871	8.81	321,734	7.59
Total	521,137	8.62	496,974	7.86	476,815	7.21

Table 4: Posterior predictive standard deviation and coefficient of variation (CoV) for the total reserve and by origin, using the exact analytic approach.

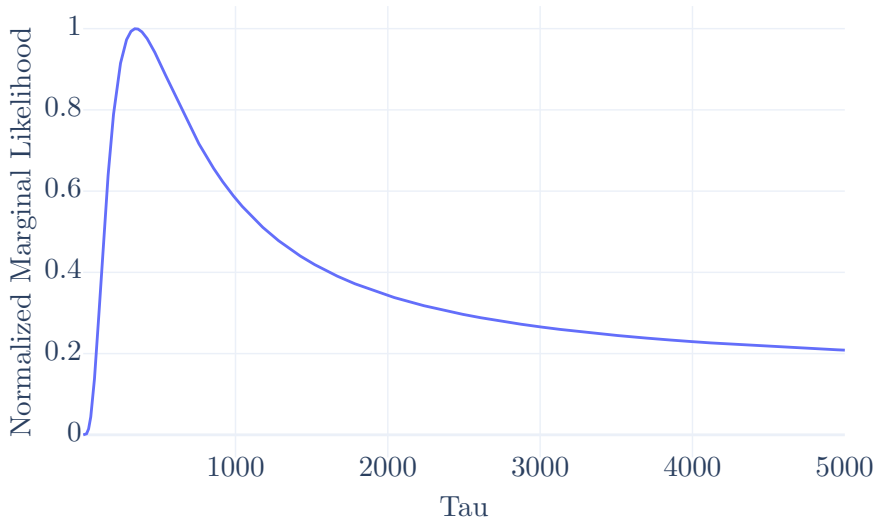


Figure 1: Normalised marginal likelihood $L_m(\tau; X)/L_m(\widehat{\tau}_{ML}; X)$ as a function of τ, with the maximum at $\widehat{\tau}_{\mathrm{ML}} = 346$

are shown in Tables 3 and 4. This scale is treated as a simple plug-in quantity for clarity. One could instead base it on the Chain Ladder fitted triangle, or iterate by re-estimating ϕ from the hierarchical fitted triangle itself, but we do not pursue those refinements here. A fully Bayesian treatment of τ via (98) remains available, but the empirical Bayes plug-in is sufficient for the present illustration. The results for the Cape Cod method are identical to those reported by Saluz [14].

Several points are worth noting. First, the total hierarchical reserve is $6,322,415$, which lies neatly between the Chain Ladder total of $6,047,061$ and the Cape Cod total of $6,617,625$. Measured on this scale, the fitted hierarchical model sits about 51.7% of the way from Cape Cod back toward Chain Ladder. The corresponding total reserve standard deviation is $496,974$, again between the Chain Ladder and Cape Cod values of $521,137$ and $476,815$ respectively.

Second, the origin-year results show the intended credibility pattern very clearly. The hierarchical model shrinks the loss ratios toward the Cape Cod estimates, but only partially. For example, for origin year 10 the posterior mean ultimate loss ratio is 64.31%, compared with 63.29% under Chain Ladder and 65.19% under Cape Cod. A similar pattern appears for origin years 8 and 9, where the low Chain Ladder loss ratios are pulled upward, but not all the way to the Cape Cod level. To quantify this more clearly, we define the *effective credibilty weight* by the relative position of the hierarchical estimate between the Cape Cod and Chain Ladder estimates, that is:

$$(R(\widehat{\tau}_{ML}) - R^{\mathrm{CC}})/(R^{\mathrm{CL}} - R^{\mathrm{CC}}) \tag{102}$$

An effective credibility weight of 100% corresponds to the Chain Ladder estimate, while 0% corresponds to the Cape Cod estimate. Figure 2 shows the effective credibility weight for each

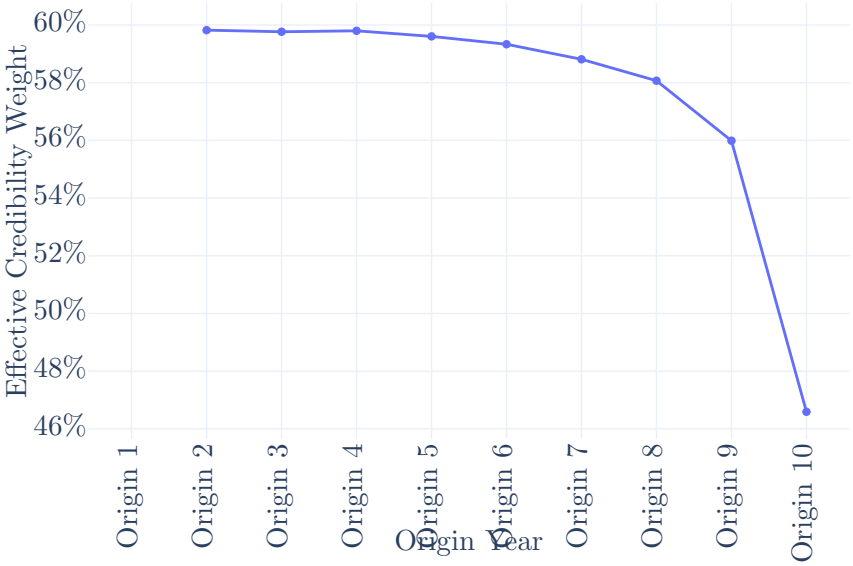


Figure 2: Effective credibility weight for each origin year under the hierarchical model. 100% corresponds to the Chain Ladder estimate, while 0% corresponds to the Cape Cod estimate.

origin year. Here it can be seen that the most recent years are more heavily weighted towards the Cape Cod estimate, with the weight shifting closer to the Chain Ladder estimate for older origin years. This is consistent with established actuarial intuition about the relative reliability of the Chain Ladder and Cape Cod methods. It is common practice, particularly for longer-tailed and/or volatile lines of business to use BF/Cape Cod estimates for more recent origin years, where the development is relatively immature and it is not clear whether the observed experience is a statistical fluctuation or evidence of a trend, whereas Chain Ladder estimates are used for older years which are more mature and the trend is more established. The actuary may even blend between Cape Cod and Chain Ladder estimates in intermediate years. The hierarchical method presented here provides a formal statistical basis for this practice, with the effective credibility weight automatically determined by the data and the exposure benchmark. While the data used for illustration in this paper is relatively short-tailed and well behaved, other examples we have tried with longer tails show this effect more significantly, with more weight given to the Cape Cod method in more recent years. A future paper will investigate the credibility weight empirically across other triangles from different lines of business and other characteristics.

The exact posterior predictive structure derived in Section 3 also makes it possible to simulate the full reserve distribution directly. Using the recursive construction based on the algorithm in Section 3.2, we obtained the predictive summaries shown in Table 5. The simulated predictive means and standard deviations are very close to the corresponding analytical values in Tables 3 and 4, as they should be. The simulation is useful as a confirmation and visualisation tool. More importantly, the simulated distributions are precisely the form needed for capital modelling applications, since quantiles and risk measures such as TVaR can be read directly from the simulated predictive distribution.

Method	Mean	SD	CoV (%)	75%	90%	95%	99%	99.5%
Chain Ladder	6,046,569	520,717	8.6118	6,385,561	6,723,529	6,930,565	7,341,631	7,497,866
Hierarchical	6,322,331	496,454	7.8524	6,645,010	6,965,277	7,163,647	7,562,709	7,715,177
Cape Cod	6,618,020	476,628	7.2020	6,927,644	7,236,033	7,427,018	7,807,918	7,954,509

Table 5: Posterior predictive summaries for the total reserve, based on 1,000,000 simulations using the simulation algorithm described in Section 3.

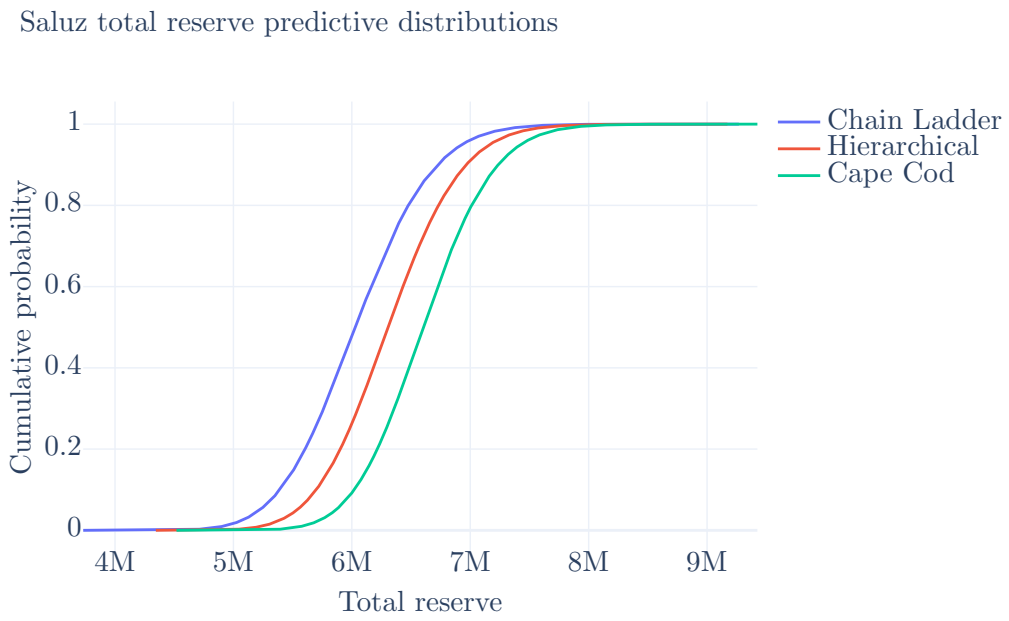


Figure 3: Posterior predictive cumulative distribution functions for the total reserve under Chain Ladder, the hierarchical credibility model and Cape Cod.

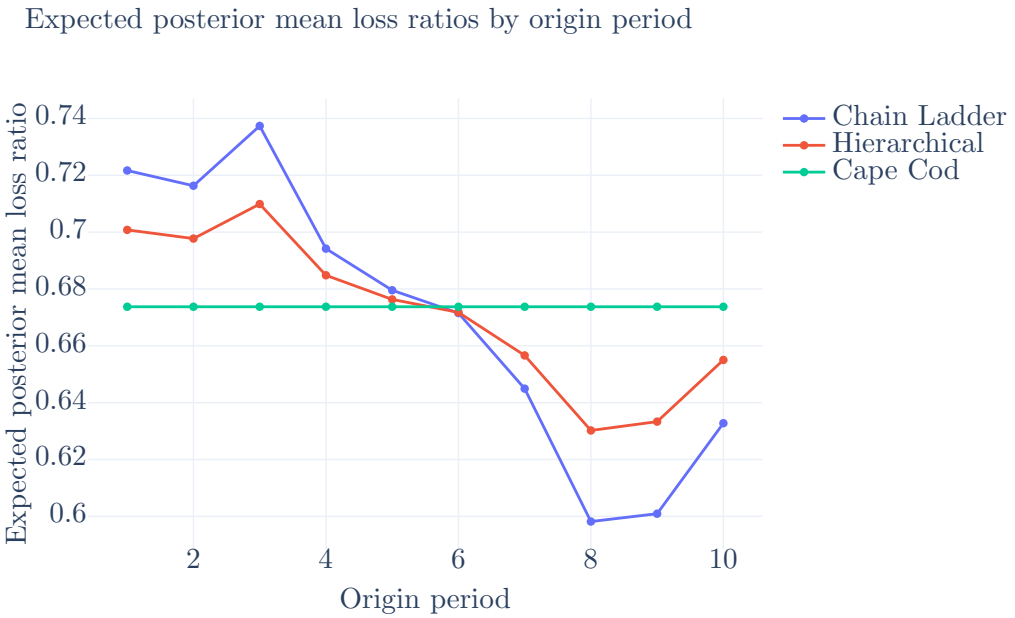


Figure 4: Expected posterior loss ratio means under Chain Ladder, the hierarchical credibility model and Cape Cod.

In Figure 4 we show the expected posterior loss ratio means, that is $\mathbb{E}[\mu_i \sum_{j=1}^{n} \beta_j / E_i | X]$ under the three models. Note that this quantity is not the same as the mean posterior ultimate loss ratio which incorporates the observed claims to date in the ultimate claims. Instead this quantity shows the effect of the underlying origin period trends. The hierarchical model again lies between the two endpoints, with a profile showing a clear shrinkage effect from the Chain Ladder toward the constant loss ratio benchmark.

6 Discussion

The results of this paper show that the ODP reserving model possesses a second exact conjugate Bayesian formulation, complementary to the development-side construction obtained under the conventional normalisation $\sum_j \beta_j = 1$. Under the alternative normalisation $\mu_1 = 1$, in which the origin period parameters can be interpreted as an index of expected claims relative to the first origin period, and after integrating out the development parameters first, the posterior on the origin side factorises into independent beta distributions. This yields a complete exact posterior predictive description of future reserve payments when prior information is placed on the origin-period parameters.

The practical value of this result is substantial. The model admits explicit closed-form reserve means, predictive variances and covariances, including a closed-form covariance matrix for origin-period future-payment totals. It supports a recursive predictive representation built only from elementary distributions, which is computationally stable and easy to implement. Just as importantly, the analytical formulas avoid the fragility and cost of high-dimensional numerical procedures. One does not need to run long MCMC chains, inspect convergence diagnostics or

accept bootstrap approximation error in order to obtain the main reserve results. It provides a single coherent ODP framework that contains Chain Ladder, Cape Cod and a continuum of intermediate credibility-weighted methods, together with a clear route to external benchmark loss-ratio information once prior structure on the overall scale is added. In the Saluz example this continuum is not merely conceptual: the fitted hierarchical reserve sits between the two classical endpoints both in mean and in predictive standard deviation, while preserving much of the origin-year structure present in Chain Ladder and damping the most unstable immature-year estimates.

This unification appears, to our knowledge, to be new. Previous Bayesian ODP papers have linked the model to familiar credibility-style reserving methods, but not through an exact closed-form origin-side predictive construction with explicit reserve-uncertainty formulas. Classical uncertainty papers for BF and Cape Cod provide important results, but do not supply a fully Bayesian ODP predictive distribution with origin-side priors in closed form. The present paper therefore helps close a genuine gap between actuarial practice and Bayesian stochastic reserving theory.

The framework also has attractive extensions. A practitioner who wishes to specify priors directly on year-on-year loss-ratio changes can do so through a prior on the relativity index μ. One may centre the benchmark relativity index on pricing-based or underwriting-based expected loss-ratio structures instead of exposures, incorporate covariate information into the prior centre, or replace the single shrinkage parameter τ by a low-dimensional hyperprior family. In that sense the benchmark relativity index is a direct mathematical bridge between pricing and reserving, with the hierarchy supplying principled partial pooling across origin periods as claims experience emerges.

A second extension developed here places a common-rate gamma prior on the development scales. In the strong exposure-centred origin-prior limit, this introduces prior credibility on the overall Cape Cod loss-ratio level. The resulting posterior mean of the common loss ratio is an exact credibility blend between the internal Cape Cod estimate and an external benchmark loss ratio. This sits naturally alongside the origin-side hierarchy, so that profile credibility and level credibility can be controlled separately within the same model.

A Gamma loss-ratio construction for the exposure-centred prior

This appendix gives one construction of the exposure-centred prior family used in Section 4.1. The construction is not needed for the main calculation, which only uses the beta prior (62), but it explains why the family has a natural loss-ratio interpretation.

Let $e_i = E_i/E_1$ and suppose that, before observing the triangle, the origin-year loss ratios are independent gamma variables

$$\Lambda_1 \sim \text{Gamma}(1+\tau, \beta), \qquad \Lambda_i \sim \text{Gamma}(\tau e_i, \beta e_i), \qquad i = 2, \ldots, n, \tag{103}$$

under the shape-rate parameterisation. For $i \geq 2$ the prior mean of Λ_i is τ/β, and its coefficient of variation is $(\tau e_i)^{-1/2}$. Thus, for origin periods $i \geq 2$, the prior loss-ratio mean is common across origin periods, the prior variance is inversely proportional to exposure, and the coefficient

of variation is inversely proportional to the square root of exposure. Origin period 1 has one additional unit of shape; this is a harmless normalising pseudo-count used to anchor the relative index under the convention $\mu_1 = 1$.

Define relative expected losses by

$$h_i := e_i \Lambda_i, \qquad i = 1, \ldots, n. \tag{104}$$

Then

$$h_1 \sim \text{Gamma}(1 + \tau, \beta), \qquad h_i \sim \text{Gamma}(\tau e_i, \beta), \qquad i = 2, \ldots, n. \tag{105}$$

Hence, after normalising by their sum,

$$\left(\frac{h_1}{\sum_{r=1}^{n} h_r}, \ldots, \frac{h_n}{\sum_{r=1}^{n} h_r} \right) \sim \text{Dir}(1 + \tau, \tau e_2, \ldots, \tau e_n). \tag{106}$$

Under the origin-side normalisation $\mu_1 = 1$,

$$\mu_i = \frac{h_i}{h_1}, \qquad i = 2, \ldots, n. \tag{107}$$

The standard stick-breaking factorisation of the Dirichlet distribution therefore gives

$$\xi_i = \frac{\sum_{r=1}^{i-1} h_r}{\sum_{r=1}^{i} h_r} \sim \text{Beta}\left(1 + \tau \sum_{r=1}^{i-1} e_r, \tau e_i \right), \qquad i = 2, \ldots, n, \tag{108}$$

independently over i, which is exactly (62). Also, since h_i is independent of h_1,

$$\mathbb{E}[\mu_i \mid \tau] = \mathbb{E}[h_i]\mathbb{E}[h_1^{-1}] = \frac{\tau e_i}{\beta} \cdot \frac{\beta}{\tau} = e_i, \qquad i = 2, \ldots, n. \tag{109}$$

This proves the exposure-centred property used in the main text.

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